

Sukhrob Ilyosbekov

Machine Learning Engineer · Computer Vision & LLM Systems

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SUMMARY

Machine-learning engineer with three published papers across computer vision, cell biology, and multimodal ML, and 9+ years building production software. Takes deep-learning and LLM work past the prototype stage: imaging models pulled into hospital workflows, an AI freight dispatcher, and multi-model pipelines running in live SaaS. Strong applied-math background, and the engineering to keep models running once they are out.

TECHNICAL SKILLS

AI/ML: PyTorch, TensorFlow, CNNs, Vision Transformers, EfficientNet, YOLO, contrastive learning, GradCAM+ +, OpenCV, LLM and agentic systems, RAG, MCP, Claude/GPT/DeepSeek APIs

Math & Modeling: Probability and statistics, linear algebra, optimization, regression, time-series forecasting, feature engineering, experiment design and A/B testing, model evaluation and error analysis

Languages: Python, C#, C++ , TypeScript, Kotlin

Data & Serving: NumPy, pandas, scikit-learn, FastAPI, PostgreSQL, MongoDB, Redis, OHIF/DICOM

Infrastructure: CUDA, mixed-precision and multi-GPU training, model serving and monitoring, Docker, Kubernetes, GitHub Actions, AWS (S3, Lambda, Cognito), Azure

Software: ASP.NET Core, Node.js, Bun, NestJS, React, Next.js, Angular

EDUCATION

Northeastern University, M.S. in Computer Science, GPA 4.0, Boston, MA

Jan 2024 – May 2026

Suffolk University, B.S. in Computer Science, Cum Laude, Boston, MA

Sep 2021 – May 2023

Honors: National Physics Olympiad, 1st Place (Uzbekistan) · Associate Member, Sigma Xi · University Coding Competition Winner, TUIT

PROFESSIONAL EXPERIENCE

Senior AI/ML Engineer | [EmTech Care Labs](#) | Portland, ME

Jan 2025 – Present

- Built the machine-learning features for a HIPAA-compliant Alzheimer's care platform, including a decline-risk model over daily activity and care-log data for roughly 1,200 residents, reaching 0.81 AUC and surfacing early signs about three weeks before the care team escalated on its own.
- Added an NLP pipeline that reads roughly 500 free-text clinical notes a day and pulls out structured details (symptoms, medications, care events) at about 0.90 F1 on a held-out set, cutting manual charting time for the care team by roughly a third.
- Stood up the ML infrastructure under HIPAA constraints: de-identified training data, nightly retraining, model serving behind the API at sub-200 ms p95, and drift monitoring. The platform cleared an outside HIPAA readiness review with no critical findings.

Machine Learning Engineer (part-time) | [AllFactors](#) | Santa Clara, CA

Oct 2021 – Dec 2024

- Built property-valuation and market-forecasting models over roughly 2M listings across 40+ metros, cutting median absolute valuation error from about 9% to 5% against the rules-based estimate the product shipped with.
- Designed the real-time feature pipeline (SignalR + SQL Server change tracking) that fed live market data to the models and pushed updated predictions to hundreds of concurrent users with sub-second latency, retraining nightly on the day's transactions.
- Productionized the models into the customer-facing dashboards and added automated tests around the data and prediction flows.

Earlier: Software Engineer at Smart Meal Service, Russia (2020 – 2021), building C#/.NET control software for self-service food kiosks and a robotic cashier. Game Developer at Pentalight Technology, Malaysia (2020 – 2021), multiplayer networking and hand tracking for a VR smart-city simulation in Unity. Freelance Developer (2019 – 2020), Python NLP and .NET applications. Game Developer at EC Dev Team, Uzbekistan (2016 – 2019).

SELECTED AI/ML PROJECTS

MorphoCLIP

[GitHub](#) | [Paper](#)

PyTorch, DINOv3, BioClinical ModernBERT, contrastive learning, CPJUMP1

- Built a text-supervised contrastive model, pairing frozen vision and language backbones (DINOv3, BioClinical ModernBERT) with trainable projection heads, that matches Cell Painting microscopy images of drug- and gene-perturbed cells to natural-language treatment descriptions, evaluated on the CPJUMP1 benchmark (51 plates, 3M+ images). Published on arXiv; collaborative work with two Northeastern co-authors, with me leading model design, training, and evaluation.

Localize, Don't Beautify

<https://arxiv.org/abs/2608.02841>

Image-editing APIs, inpainting, ArcFace, prompt engineering

- Studied why commercial image-editing APIs over-edit the face during cosmetic-surgery preview generation and how much of that can be fixed client-side without touching the model; tested prompt-only, masked-compositing, and model-based inpainting across six editors, and found simple masking beat model-based inpainting on localization accuracy. Published on arXiv.

MelanomaNet

[GitHub](#) | [Paper](#)

PyTorch, EfficientNet V2, GradCAM++ , OpenCV

- Trained a deep-learning model to classify skin lesions across nine diagnostic categories on a standard public dermatology dataset of 25K+ images, reaching 0.86 weighted F1.
- The research contribution is interpretability: the model produces heatmaps of what it looked at, decomposed along the same ABCDE criteria (Asymmetry, Border, Color, Diameter, Evolution) dermatologists already use, so a clinician can follow and trust the reasoning rather than getting an opaque score. Published on arXiv.

Med Image Scanner

<https://github.com/suxrobgm/med-image-scanner>

FastAPI, PyTorch, OpenCV, OHIF Viewer, DICOM, Next.js

- HIPAA-compliant platform that pulls X-ray, CT, and MRI scans directly from hospital imaging systems and runs PyTorch detection models on them, flagging pneumonia on chest X-rays and intracranial hemorrhage on head CTs.
- Renders model predictions as overlays inside a real radiology viewer (OHIF), and handles patient data safely throughout, with on-the-fly de-identification, audit logging, and role-based access.

LogisticsX

<https://logisticsx.app> | [GitHub](#)

Claude API, MCP, .NET 10, EF Core, SignalR, Angular 21, Kotlin KMP, PostgreSQL

- Built an AI dispatcher that matches freight loads to trucks, checks federal driver hours-of-service rules, and plans multi-stop routes on its own, turning a manual 15-minute decision into a near-instant one. Runs on the Claude API through a custom tool-use agent with its own set of tools.
- Shipped it as a real product: web portals for dispatchers, customers, and brokers, a cross-platform driver app, and a multi-tenant backend integrated with the major freight load boards and tracking providers.

Bookshelf Scanner

<https://github.com/suxrobgm/bookshelf-scanner>

YOLO, Moondream2 VLM, FastAPI, Angular

- Two-stage computer-vision pipeline: a YOLO segmentation model isolates each individual book spine from a shelf photo, then a vision-language model reads the title and author off each spine, wrapped in a web app for correcting misreads and exporting the library.

OGStack

<https://ogstack.dev> | [GitHub](#)

Next.js 16, ElysiaJS, Bun, Satori, Cloudflare R2, Stripe

- Built a multi-model AI layer that routes work across Claude, GPT, and DeepSeek to read a web page and match its branding, then generates an on-brand social-share image, including AI-generated backgrounds. Runs as a SaaS serving cached images from a global CDN; it generated 10K+ images during the beta.